

Solar-Powered Bird Scarer

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Presented here is a solution to scare off birds from your farm house, warehouse or fruit garden. Usually birds dislike a rasping sound of around 200 Hz and will avoid that environment. The circuit takes advantage of this fact and generates sound vibrations at a frequency of 211.5 Hz. It is fully powered by a solar panel and there is no need for a battery or AC power.

PARTS LIST

Semiconductors:

- IC1 - 7555 timer
D1 - 1N4007 rectifier diode

Resistors (all ¼-watt, ±5% carbon):

- R1 - 68 kilo-ohm
R2 - 100 ohm

Capacitors:

- C1 - 470µF, 25V electrolytic
C2 - 0.1µF ceramic disk
C3 - 0.01µF ceramic disk

Miscellaneous:

- SP1 - 2-pin mating connector
PZ1 - 6V, 300mA solar panel
S1 - Piezo-buzzer
S1 - On/off switch

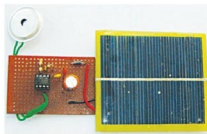


Fig. 1: Author's prototype of the solar-powered bird scarer

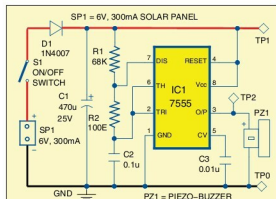


Fig. 2: The circuit of the solar-powered bird scarer



The bird scarer is portable and can be placed anywhere, but if it remains in the same place continuously for three to four weeks, the birds may grow accustomed to the sound—so it has to be used intermittently. It is found effective within a radius of 2-3 metres, and is an ideal choice for places where food grains are being sun dried. Fig. 1 shows the author's prototype.

Circuit and working

The circuit of the solar-powered bird scarer is shown in Fig. 2. It is built around popular timer 7555 (IC1), diode 1N4007 (D1), a 6V, 300mA solar panel (SP1) and some commonly used components.

The circuit is fully powered from a 6V, 300mA mini solar panel. In bright sunlight, the solar panel generates around 10V, which powers the circuit via diode D1. Capacitor C1 acts as a buffer to provide steady voltage to the circuit.

IC1 is wired as an astable multivibrator to generate 211.5Hz output pulses to activate the piezo-buzzer. In astable mode, its output voltage alternates on a continuous basis. The period of the astable output is the time it takes for the on/off cycle to repeat itself, whilst the duty cycle is the percentage of time the output is on. The frequency/period and the duty cycle of the astable multivibrator is determined by resistors R1, R2 and the capacitor C2. So, with the given values of the timing resistors and capacitors, output pulses from IC1 will be at 211.5 Hz. The output from IC1 makes the piezo-buzzer ring.

How the circuit works is

Test Points

Test point	Details
TP0	0V, GND
TP1	Solar panel voltage
TP2	Around 211.5 Hz frequency

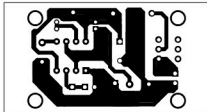


Fig. 3: A single-side PCB for the solar-powered bird scarer, shown in actual size

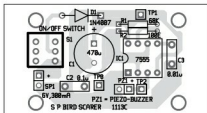


Fig. 4: Component layout for the PCB

simple. In sunlight, the piezo-buzzer rings and scares off the birds from the surroundings if switch S1 is closed. The output frequency can be adjusted by varying the values of the timing components.

Construction and testing

A single-side PCB for the solar-powered bird scarer is shown in actual size in Fig. 3 and its component layout is shown in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable plastic case. Connect the solar panel to the PCB at SP1. The circuit requires sufficient sunlight to power it, so place the unit in direct sunlight. If a particular bird population does not respond to this frequency, replace R1 with a 100k preset, and adjust the frequency through trial and error.

To test if the circuit is functioning properly, check solar voltage at TP1 with respect to TP0. The output frequency can be observed at TP2, using an oscilloscope. ●

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